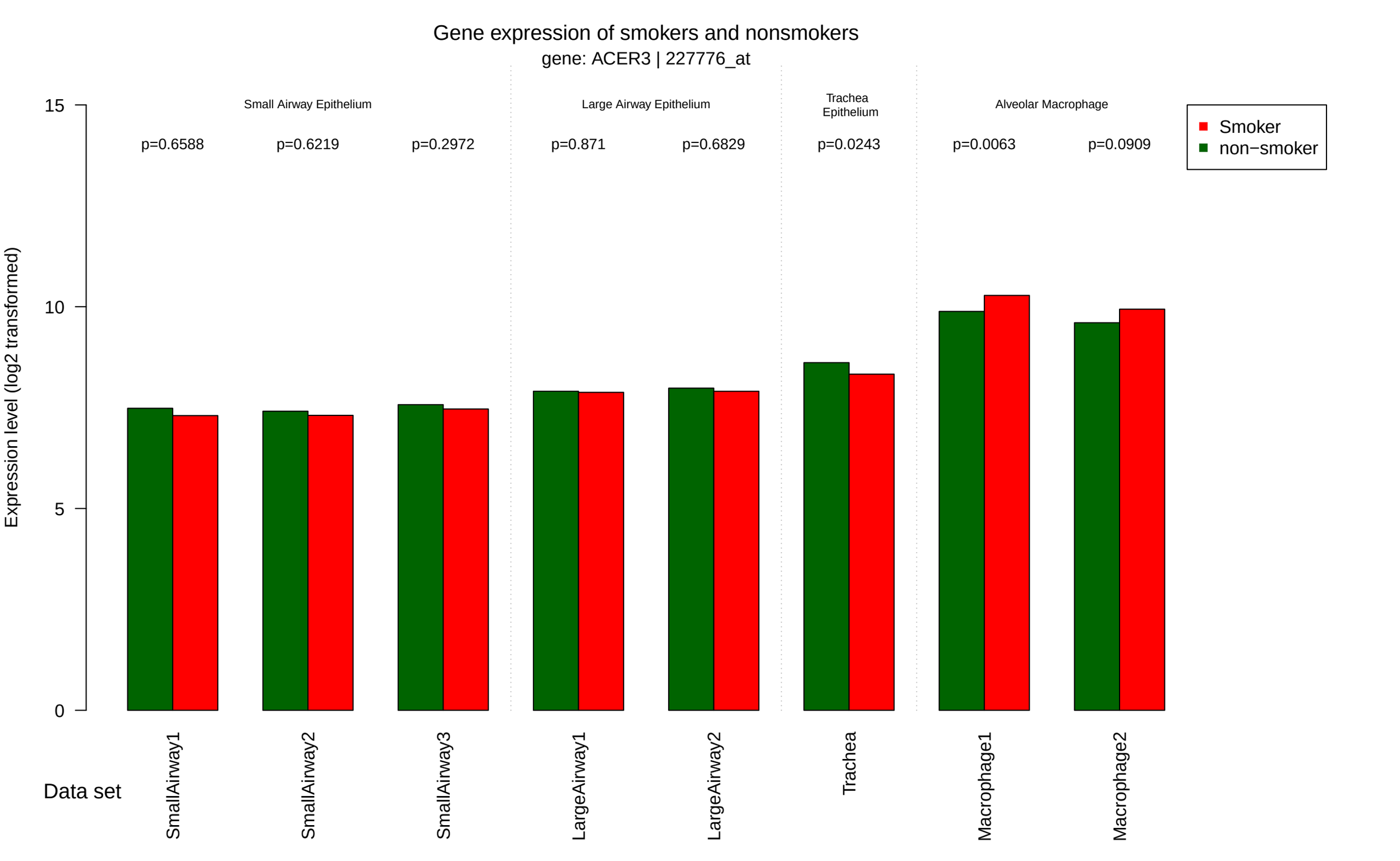
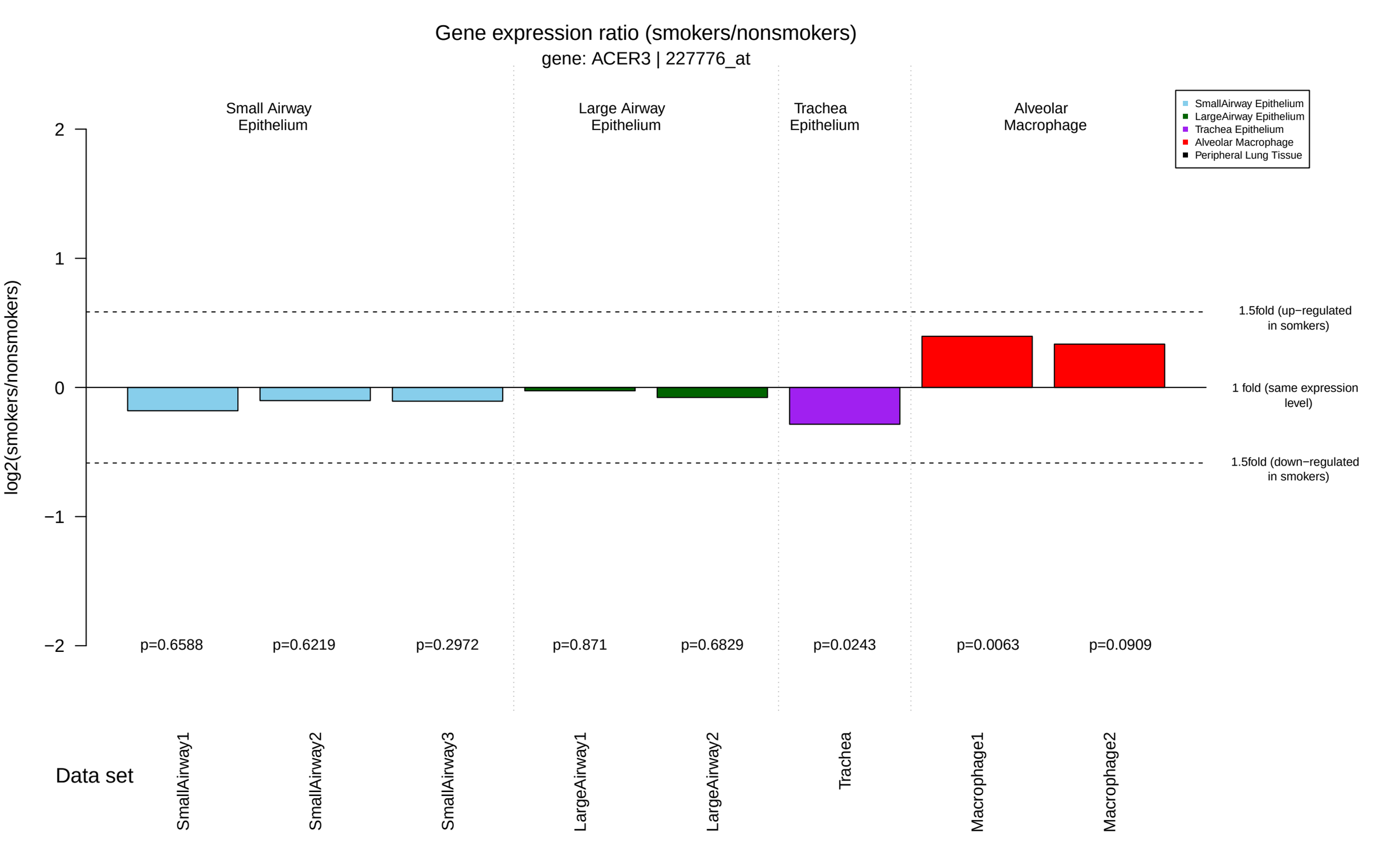
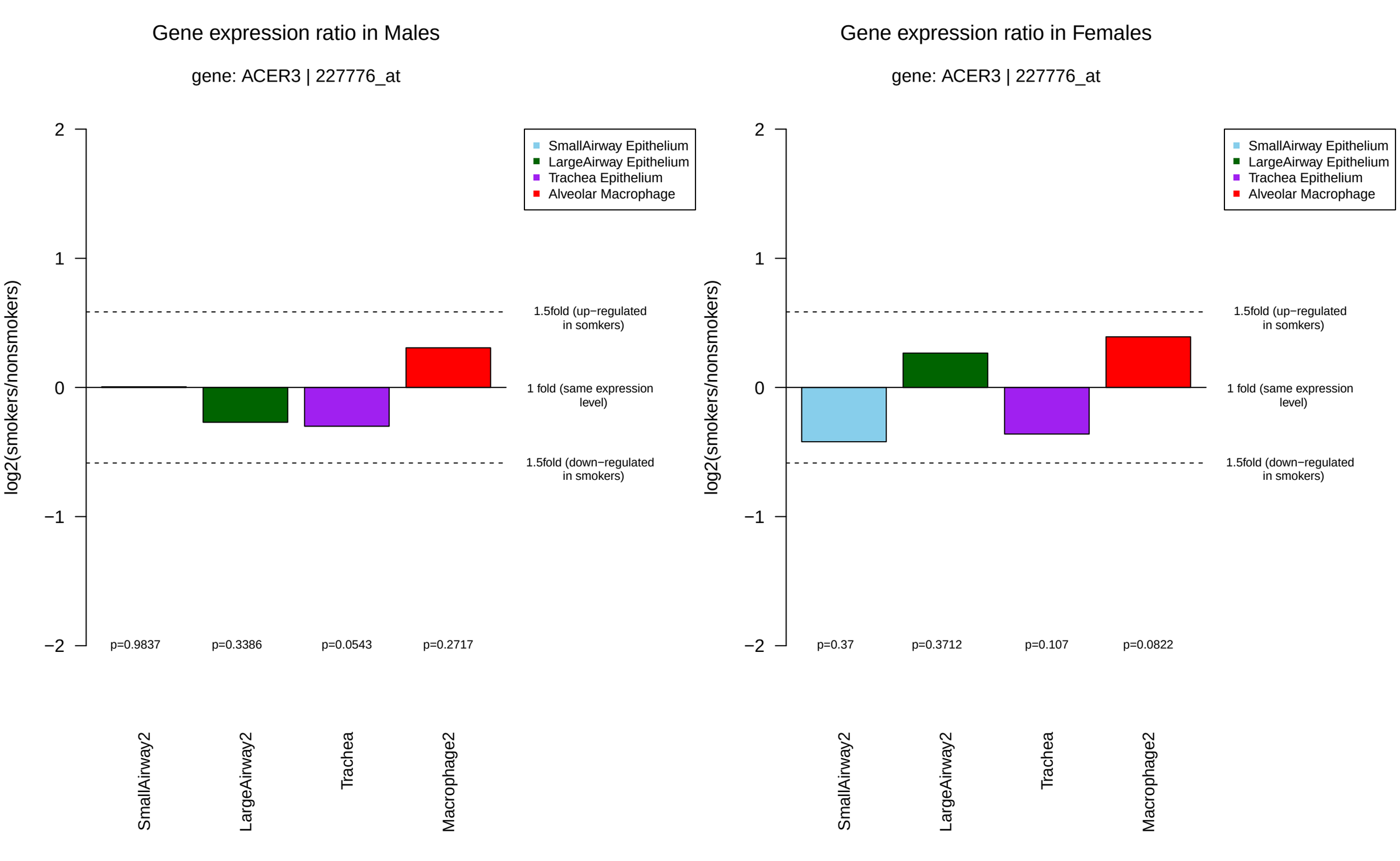
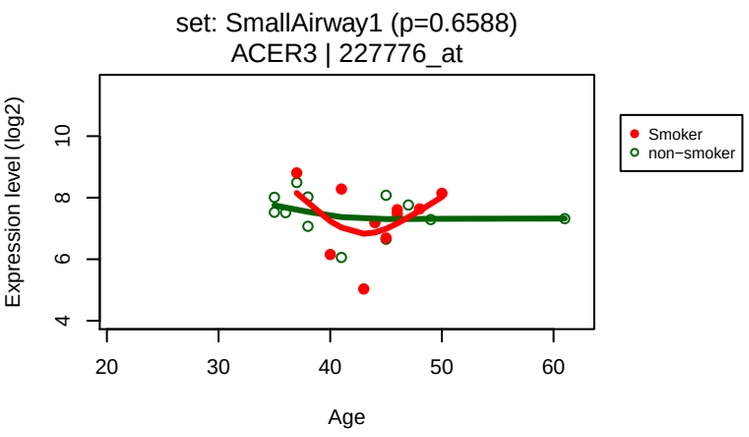
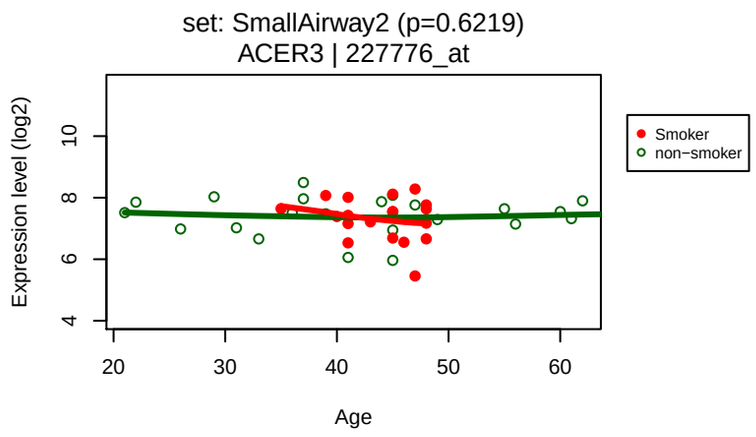
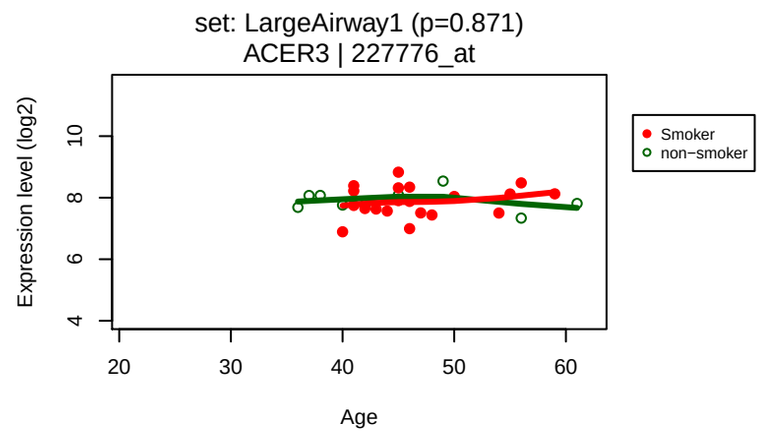

Gene expression of smokers and nonsmokers
gene: ACER3 | 227776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6588
p=0.6219
p=0.2972
p=0.871
p=0.6829
p=0.0243
p=0.0063
p=0.0909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

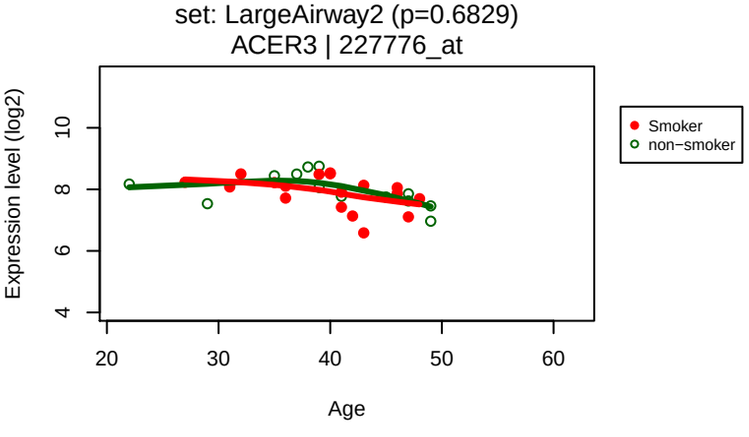

Gene expression ratio (smokers/nonsmokers)
gene: ACER3 | 227776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6588
p=0.6219
p=0.2972
p=0.871
p=0.6829
p=0.0243
p=0.0063
p=0.0909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

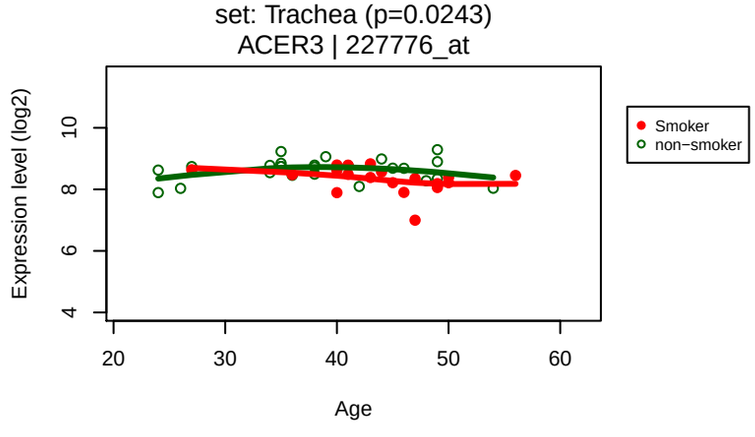

Gene expression ratio in Males
gene: ACER3 | 227776_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9837
p=0.3386
p=0.0543
p=0.2717
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER3 | 227776_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.37
p=0.3712
p=0.107
p=0.0822
SmallAirway2
LargeAirway2
Trachea
Macrophage2

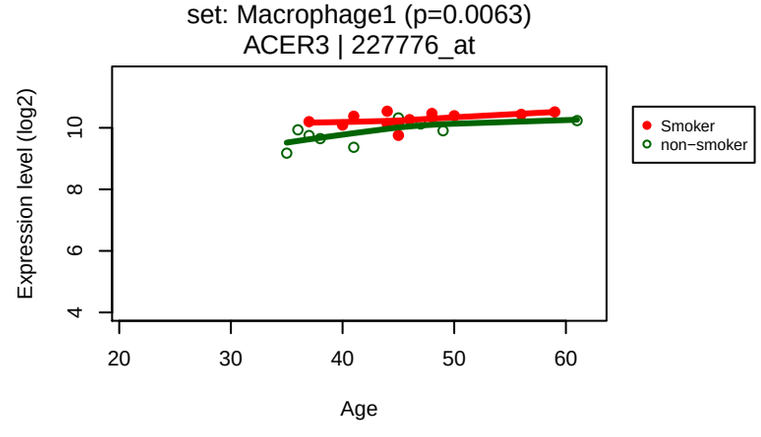

set: SmallAirway1 (p=0.6588)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker

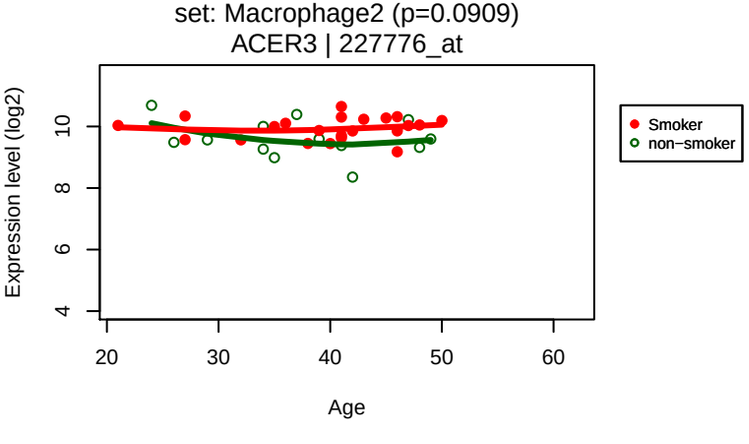

set: SmallAirway2 (p=0.6219)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker

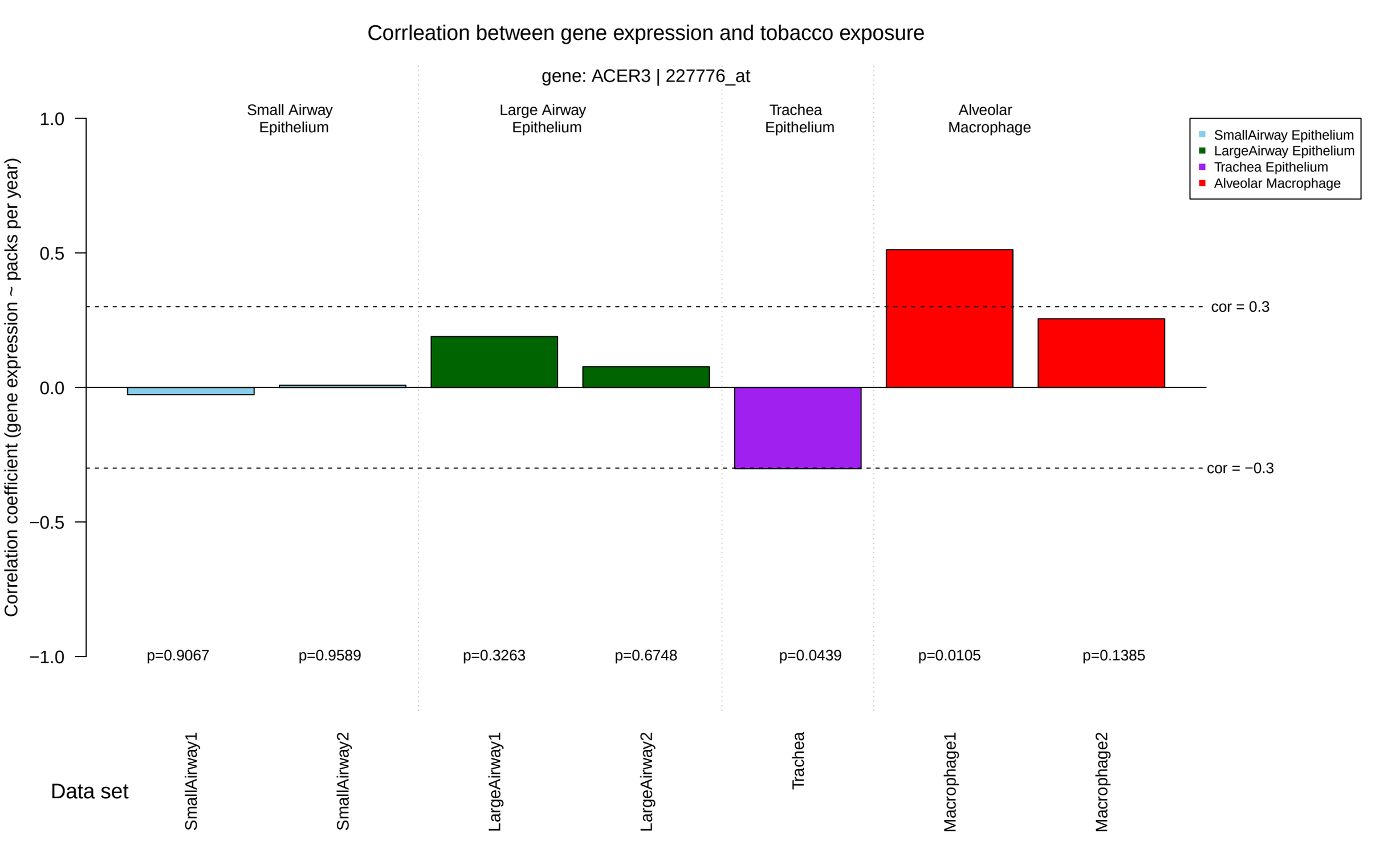

set: LargeAirway1 (p=0.871)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6829)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0243)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0063)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0909)
ACER3 | 227776_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER3 | 227776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9067
p=0.9589
p=0.3263
p=0.6748
p=0.0439
p=0.0105
p=0.1385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2